



Incident report analysis

Instructions

Create a plan to improve your company's network security, following the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF)

Summary	The organization recently experienced a network outage that lasted approximately two hours before services were restored. During the incident, several of the organization's services became unavailable due to a flood of ICMP packets targeting the company's internal network. The incident response team promptly initiated containment and recovery procedures, after which the cybersecurity team conducted an investigation. The investigation revealed that a malicious actor exploited a misconfigured firewall to send excessive ICMP traffic into the network. As a result, the attacker successfully launched a Denial of Service (DoS) attack that disrupted critical network services.
Identify	The incident management team audited the systems, devices, and access policies involved in the attack to identify gaps in the security. They found that the company's internal network is being flooded by ICMP packets. Upon initial review, it appears that the malicious actor had sent a flood of ICMP pings into the network through an unconfigured firewall.
Protect	The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring the critical ones.
Detect	To detect new DoS attacks in the future, the security team will use: - A new firewall rule to limit the rate of incoming ICMP packets - Source IP address verification on the firewall to check for spoofed IP

	addresses on incoming ICMP packets - Network monitoring software to detect abnormal traffic patterns - An IDS/IPS system to filter out some of the ICMP traffic based on suspicious characteristics
Respond	To respond to future ICMP DoS attacks, the organization should establish an incident response plan that includes detecting abnormal ICMP traffic, activating response teams, and implementing predefined containment measures. Regular testing and updating of the plan will ensure readiness and minimize service disruption during attacks.
Recover	After the attack, the team restored normal communication across the organization's internal network and verified that all affected services were functioning correctly. For future attacks, network configurations and critical system settings should be backed up and documented to support rapid recovery. Continuous monitoring should also be maintained to ensure that malicious traffic does not reoccur after the incident has been resolved.
